

Waraitip Laosangprateep

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BKK-EEC · Chonburi-Rayong-Samut prakarn · Available from August 2026



PROFILE

Medical Engineering master's student with a Chemical Engineering foundation and hands-on exposure to GMP-aligned manufacturing, quality control practices, and biomedical laboratory research. Experienced in precision instrumentation, regulatory frameworks, and process documentation through internships in industrial food manufacturing and international research institutions. Seeking a Quality Assurance, Regulatory Compliance, or Process Engineering role within a medical device, pharmaceutical, or consumer health company, where technical rigour and a quality-focused mindset support safe, compliant product manufacturing.

EDUCATION

- Master of Medical Engineering** · **University of Auckland** · New Zealand Mar 2025 – Jun 2026
- Thesis: Mechanical characterisation of growth plate structures using microindentation techniques
 - Coursework: Advanced Functional Materials · Medical Device & Technology Development · Advanced Drug Disposition & Kinetics · Advanced Imaging Technologies · Engineering Project Management
- Bachelor of Engineering – Chemical Engineering** · **Thammasat University (SIIT)** · Thailand Aug 2020 – May 2024
- GPA: 3.22 / 4.00 · International Programme, full English instruction
 - Senior Project: Theoretical study on butylone inclusion complexes with β -cyclodextrin — presented at PACCON 2024
 - Core coursework: Chemical Reaction Engineering · Process Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics

RESEARCH & INDUSTRY EXPERIENCE

- Master's Thesis Researcher** · **University of Auckland** · New Zealand Mar 2025 – Present
- Performed nanoindentation and microindentation mechanical testing on biological tissue samples across growth plate sub-structures
 - Applied critical analysis to reconcile conflicting experimental data and form evidence-based conclusions
 - Developed expertise in precision instrumentation, sample preparation, and quantitative data interpretation
- Research Intern – SWCNTs Defect Characterisation** · **Academia Sinica (IAMS-IIP)** · Taiwan Jun – Jul 2023
- Conducted independent experiments on Single-Walled Carbon Nanotubes using absorbance and fluorescence spectroscopy for cancer diagnostic applications
 - Designed and executed DOE-style experiments; analysed spectral data and contributed to peer scientific reports
 - Gained proficiency in nanomaterial characterisation, precision instrumentation, lab safety protocols, and scientific communication

TECHNICAL SKILLS

Quality & Regulatory: GMP manufacturing; quality control checkpoints; factory documentation & compliance; CAPA methodology awareness; quality system principles

Medical Device Standards: ISO 13485 (medical device QMS); ISO 14971 (risk management); IEC 60601 (medical electrical equipment safety); IEC 62366 (usability engineering for medical devices); EU MDR / CE marking awareness — coursework exposure

Medical Device: Medical device design principles; FMEA (Design & Process); FTA (Fault Tree Analysis); nanoindentation of biological tissue; CAD modelling (beginner); product validation concepts (IQ/OQ/PQ basics)

Biomedical Lab: Absorbance & fluorescence spectroscopy; nanoindentation systems; optical & electronic instrumentation; sample preparation & analysis

Regulatory Compliance: Biomedical waste management; chemical disinfection; pathogen risk assessment; decontamination technologies; lab biosafety (PHAC certified, 10+ certs)

Process Engineering: Chemical & process engineering fundamentals; thermodynamics; heat & mass transfer; reaction engineering; GMP manufacturing principles

Data & Computing: Python (data analysis, visualisation, supervised ML); statistical analysis; MS Project; Origin Lab; MS Office Suite

ADDITIONAL EXPERIENCE

- Customer Service** · Auckland, NZ · **Lily's Collection · Blue Elephant · Khao San Eatery · The Coffee Club** Aug 2024 – Jan 2026
- Developed stakeholder communication, cross-cultural teamwork, and operational reliability skills transferable to customer-site engineering contexts

VOLUNTEER & LEADERSHIP

- Narrator & Lab Demonstrator** · **Thammasat University – SIIT Open Days** · Thailand Jul 2022 – May 2024
- Facilitated chemical process demonstrations for prospective students; contributed to increased enrolment through clear technical communication

CERTIFICATIONS & ACHIEVEMENTS

- Public Health Agency of Canada Certifications (10+): Biomedical Waste Management · Chemical Disinfectant · Pathogen Risk Assessment · Decontamination Technologies & Lab Biosafety
- Poster Presentation Certificate – PACCON 2024 (Pure and Applied Chemistry International Conference)
- Internship Certificate – Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan

LANGUAGES

Thai (Native) · **English** (IELTS 6.5 · TOEIC 830 — Professional) · **Japanese** (Elementary)